

Topic 5: Arms Race

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Overview



Scenario: Purple (P) and Green (G) are two countries competing in an arms race.

Research Question: What happens to the long-term spending of each country?

Key Factors:

- More inclined to increase defence spending as the other's rises
- Less inclined to increase spending as its own spending rises
- Does public sentiment affect defence spending?
- Does limiting spending affect a country's position?

Motivation: It helps predict military escalation between countries, which can inform policies to promote peace and prevent conflict.



Base Model

Model Equations

The equations for the base model are:

$$\frac{\partial p}{\partial t} = ag(t) - bp(t)$$

$$\frac{\partial g}{\partial t} = cp(t) - dg(t)$$

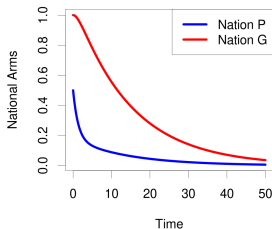
- Feasibility Condition: $(p(t), g(t) \geq 0)$
- a, c : Mutual Fear ($a, c > 0$)
- b, d : Expenditure ($b, d > 0$)

Long Term Behavior (Base Model)



$$bd > ac$$

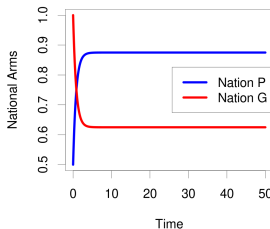
$$a=0.1, b=0.7, c=0.2, d=0.1$$



Complete mutual
disarmament

$$bd = ac$$

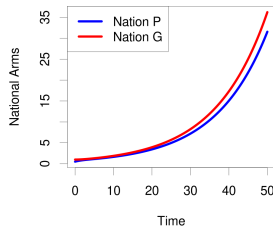
$$a=0.7, b=0.5, c=0.5, d=0.7$$



Stabilized spending

$$bd < ac$$

$$a=0.5, b=0.5, c=0.2, d=0.1$$



Perpetual arms race

Public Sentiment (Trust and Hostility)



Model Equations

The equations after including public sentiment are:

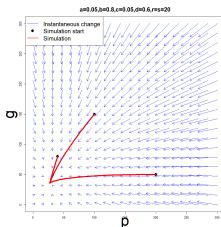
$$\frac{\partial p}{\partial t} = ag(t) - bp(t) + r$$

$$\frac{\partial g}{\partial t} = cp(t) - dg(t) + s$$

■ $(r, s) > 0$ for trust and $(r, s) < 0$ for hostility

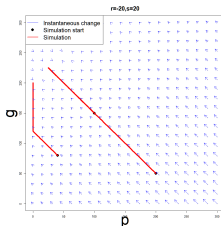
Long Term Behavior (Public Sentiment)

$$bd > ac \text{ or } bd = ac, r < 0, s < 0$$



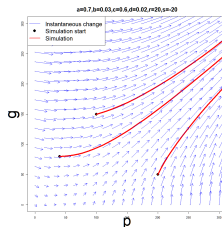
Complete mutual disarmament

$$bd = ac, r < 0, s > 0$$



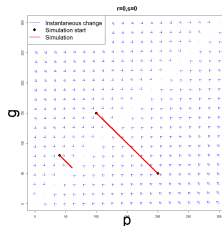
One nation grows, other nation shrinks

$$bd < ac \text{ or } bd = ac, r \geq 0, s > 0$$



Perpetual arms race

$$bd = ac, r = 0, s = 0$$



Nations never reach equilibrium

Mutual Logistic Fear



Limitation of the Current Model: Each country has a limited amount of resources at a given time.

Solution: Introduce logistic growth.

Model Equations

The equations after adding logistic growth to account of limited resources are:

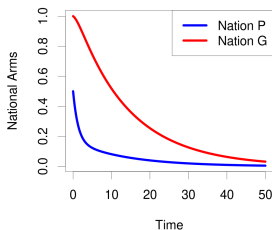
$$\begin{aligned}\frac{\partial p}{\partial t} &= a\left(1 - \frac{p(t)}{K_p}\right)g(t) - bp(t) + r \\ \frac{\partial g}{\partial t} &= c\left(1 - \frac{g(t)}{K_g}\right)p(t) - dg(t) + s\end{aligned}$$

Long Term Behavior (Mutual Logistic Fear)



$$bd > ac$$

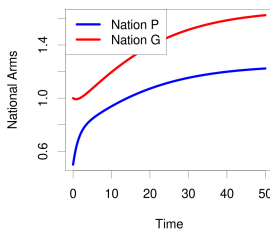
$$a=0.1, b=0.7, c=0.2, d=0.1, K_p=5, K_g=5$$



Complete mutual
disarmament

$$bd = ac$$

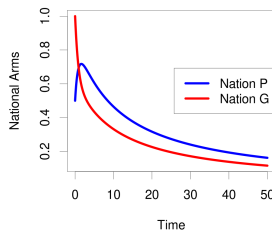
$$a=0.5, b=0.5, c=0.2, d=0.1, K_p=5, K_g=5$$



Stabilized spending

$$bd < ac$$

$$a=0.7, b=0.5, c=0.5, d=0.7, K_p=5, K_g=5$$



Perpetual arms race

Thank you for listening!



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